Harpreet Singh

Neurotechnology / BCI Engineer

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Summary

Neurotechnology and BCI research engineer combining computational behavioural neuroscience, embedded systems, EEG/fNIRS acquisition, synchronized behavioural experiments, Python analysis, LSL/LabRecorder workflows, and FPGA-based neuromorphic research. Current PhD student and Research Assistant at the University of Lethbridge building practical tools for brain-care, behavioural neuroscience, and neural decision modeling.

Core Skills

- EEG, fNIRS, BCI, computational behavioural neuroscience, neural data acquisition, behavioural task markers, XDF sessions
- ADS1299, ADS8688, DAC8565, Raspberry Pi, SPI, GPIO, TDM MUX, watchdog PWM, live plotting, CSV logging
- Lab Streaming Layer, LSL markers, LabRecorder, CGX EEG, UDP marker bridge, stream probing, session manifests
- Python, NumPy, Pandas, Matplotlib, scikit-learn, PyTorch, TensorFlow/Keras, biomedical signal analysis
- SNN, QSNN, QiSNN, rodent raster, neural population dynamics, FPGA acceleration, MicroBlaze, BRAM, UART diagnostics

Professional Experience

Research Assistant, Core-Hub Neuroengineering Solutions | *University of Lethbridge | Apr 2025 - Present*

- Build BCI and neurotechnology research systems for EEG/fNIRS acquisition, behavioural task synchronization, embedded device control, and neural data analysis.
- Developed 8-channel EEG + 8-channel fNIRS Raspberry Pi acquisition software using ADS1299, ADS8688, DAC8565, TDM MUX sequencing, live plotting, and offline CSV analysis.
- Created Sheeg software for CGX EEG, behavioural game execution, UDP-to-LSL event markers, LabRecorder control, XDF recording, stream probes, manifests, and session summaries.

Research Assistant | *University of Regina | Jan 2021 - Apr 2024*

- Researched uncertain-reasoning intrusion detection for DoS/DDoS detection using Bayesian Networks, Markov Networks, Zeek, Wireshark, feature engineering, and Python ML workflows.
- Published SECRYPT 2024 work on uncertain-reasoning IDS methods and presented findings to an international cybersecurity research audience.
- Built ML prototypes including a Bayesian liver-disorder diagnostic model with 85% accuracy and Punjabi BERT/NLP classifiers for 100K+ news articles with 92% classification accuracy.

Independent Embedded Systems Developer | *Self-Employed | Jun 2024 - Mar 2025*

- Designed a custom STM32F405RGT6 MicroPython development board, including multilayer PCB design, firmware flashing, sensor/actuator interfaces, power management, and hardware/software debugging.

Senior Software Engineer | *Planetcast Media Services Ltd. | Dec 2018 - Nov 2020*

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature acquisition platform.
- Designed schematics and PCBs, selected components, collaborated on cabinet/mechanical integration, and developed C firmware with Ethernet, HTTP/LWIP, alarm, and control interfaces.

Embedded Software Engineer / R&D Engineer | *Spark Eighteen Pvt. Ltd.; Exicom Tele-systems Ltd. | Feb 2016 - Dec 2018*

- Developed STM32/AVR embedded systems for LiFi control, telecom power plants, DC interface cards, and automatic test equipment.
- Designed 15-20 W isolated SMPS sections with 78-81% measured efficiency and performed board bring-up across CAN, SPI Flash, I2C EEPROM, Ethernet/LWIP, and GUI test automation.

Selected Projects

8eeg8fNIRS

- Integrated ADS1299 EEG, ADS8688 fNIRS analog acquisition, DAC8565 output control, watchdog PWM, TDM MUX sequencing, and Raspberry Pi SPI/GPIO concurrency into live acquisition scripts.

Sheeg

- Built synchronized EEG + behavioural experiment orchestration with CGX Acquisition, UDP-to-LSL marker bridge, LSL stream probing, LabRecorder remote control, XDF recording, and session metadata.

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- Prepared rodent decision decoding pipeline that converts neural population raster windows into 12 x 196 temporal features for no-lick, left-lick, and right-lick prediction experiments.

Publication

- Harpreet Singh et al., "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection," SECRYPT 2024, ISBN 978-989-758-709-2, ISSN 2184-7711, pp. 771-776.

Education

- PhD in Computational Behavioural Neuroscience, University of Lethbridge - In progress
- M.Sc. Computer Science, University of Regina - 2024
- Bachelor's equivalent in Electronics & Telecommunication Engineering, AMIETE/IETE - 2012